PASTE: Physics-Aware Scattering Topology Embedding Framework for SAR Object Detection

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***Abstract*—Deep synthetic aperture radar (SAR) object detectors often inherit optical-image paradigms and consequently underuse the sparse, discrete electromagnetic scattering structure of targets. Existing scattering-enhancement methods commonly rely on amplitude statistics or costly frequency-domain processing, making it difficult to incorporate physically meaningful structure into modern detectors. We propose physics-aware scattering topology embedding (PASTE), a closed-loop framework that integrates scattering topology priors into SAR object detection. Scattering keypoint automatic annotation (SKAA) automatically generates attributed scattering center (ASC)-based keypoint annotations, scattering topology injection module (STIM) injects the learned topology into multiscale features, and scattering prior supervision strategy (SPSS) uses Gaussian soft targets to supervise the scattering-map branch. Across seven convolutional neural network (CNN)-, Transformer-, and Mamba-based detectors, PASTE yields relative mean average precision (mAP)@0.5 gains of 1.3%–11.3%. Visualizations further show that the learned scattering representations distinguish target and background regions, improving interpretability together with detection accuracy. In addition, parameter ablation experiments and robustness perturbation experiments are also conducted to verify the practicality of PASTE.**

***Index Terms*—Attributed scattering center (ASC) model, physics prior fusion, physics-aware synthetic aperture radar (SAR) object detection, scattering topology prior.**

I. Introduction

With the rapid proliferation of synthetic aperture radar (SAR) data and the advancement of deep learning (DL), SAR object detection has increasingly relied on DL-based frameworks. While DL-based detectors have achieved remarkable success in optical imagery [1], [2], [3], directly transferring these paradigms to SAR remains suboptimal due to fundamental differences in imaging mechanisms. As shown in Fig. 1, an aircraft in an optical image is characterized by continuous geometric structures and coherent textures; in an SAR image, the same target is represented by sparse, discrete scattering centers governed by electromagnetic interactions [4], [5], [6]. The geometric structure of a target is only implicitly represented through the spatial configuration of these scattering centers [7]. However, most existing SAR detection methods are adapted from optical frameworks, including two-stage detectors (e.g., Faster/Cascade R-convolutional neural network (CNN) bib:8, [9]), one-stage methods (e.g., FCOS [10], YOLO series [11], [12]), and Transformer-based architectures [13], [14]. These approaches implicitly assume that targets can be modeled as local texture or intensity distributions. Such representations are not fully consistent with the scattering-based nature of SAR imaging, where discriminative information is determined by a limited number of dominant scatterers and their geometric configurations. This mismatch often leads to degraded robustness and interpretability, especially under limited training data conditions [15].

To alleviate this issue, several studies attempt to incorporate SAR-specific information such as amplitude, phase, polarization, or complex-valued representations into SAR learning



Fig. 1. (Top) Optical and (bottom) SAR images of the same terminal area in Changi Airport. The right side of the figure highlights the same aircraft target using a red region and yellow dashed lines, respectively. The aircraft appears as a continuous region in the optical image but as discrete bright patches in the SAR image.

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pipelines [16], [17], [18], [19]. Many of these methods rely on feature-level fusion strategies, including channel concatenation, attention-based fusion, or multibranch architectures [20], [21], [22], [23], [24], [25]. Although effective to some extent, these approaches primarily operate on signal-level representations and fail to explicitly model the structured and discontinuous topology formed by scattering centers. Treating sparse scattering responses as continuous features may introduce background interference and limit the ability to distinguish true targets from pseudo-bright clutter.

Another line of research explores structural representations using keypoints or topology-aware modeling. These methods identify salient points or structural anchors to enhance feature representation. However, most existing approaches derive such keypoints from image statistics (e.g., intensity or saliency), making them heuristic and lacking physical interpretability. In contrast, scattering centers derived from electromagnetic models, such as the attributed scattering center (ASC) model, provide a physically grounded description of target structure. Nevertheless, integrating ASC-based representations into DL frameworks remains challenging due to the lack of scalable annotation mechanisms and effective integration strategies.

To address these limitations, this article proposes a physics-aware scattering topology embedding (PASTE) framework, which introduces a closed-loop paradigm for integrating scattering topology priors into SAR detection. Specifically, PASTE constructs a unified pipeline consisting of scattering topology prior generation, topology injection, and prior-guided supervision. First, an ASC-based scattering keypoint generation and automatic annotation method, termed scattering keypoint automatic annotation (SKAA), is designed to extract physically consistent keypoints and form scattering topology representations without manual labeling. Then, a scattering topology injection module (STIM) incorporates these scattering topology priors into multiscale feature learning. Furthermore, a scattering prior supervision strategy (SPSS) imposes Gaussian-based soft constraints to align network predictions with the spatial distribution of scattering centers.

The main contributions of this article can be summarized as follows.

1) We propose the PASTE framework that connects ASC-derived scattering topology priors with DL-based SAR detectors through prior generation, feature injection, and prior-guided supervision.
2) We develop SKAA, which decouples target scattering regions, estimates ASC positions, and clusters them into fixed-cardinality scattering keypoint annotations without manual keypoint labeling.
3) We introduce STIM and SPSS. STIM injects predicted scattering maps into feature pyramid network (FPN) features, while SPSS uses keypoint-centered Gaussian targets to align the map prediction branch with the scattering topology prior during training.
4) Experiments across multiple detector families and imaging modes show consistent improvements. Ablations further show that ASC-derived keypoints provide a more effective scattering topology prior than amplitude-based keypoints.

The remainder of this article is organized as follows. Section II reviews SAR object detection and scattering topology priors. Section III details the proposed method and its three components. Section IV presents the experimental setup, comparisons, ablations, and qualitative analyses. Section V concludes this article and discusses future work.

II. Related Work

A. Traditional Methods for SAR Object Detection

Traditional SAR detection methods are mainly based on statistical modeling and image processing techniques, such as CFAR and SVM [26], [27]. Among them, CFAR estimates local clutter distributions and applies adaptive thresholding to detect targets. Despite their effectiveness in simple scenarios, these methods rely on limited statistical assumptions and lack the ability to capture high-level semantic and structural information, resulting in poor performance in complex environments.

Subsequent studies introduced more advanced models, including mixture distribution-based methods [28] and ASC-based template matching approaches [29], [30]. Although these methods incorporate certain physical insights, they are typically designed for specific scenarios and lack scalability and generalization capability.

B. DL-Based Methods and Structural Representations

DL-based methods have significantly advanced SAR object detection by leveraging CNNs and Transformer architectures [31], [32], [33], [34]. These approaches improve feature representation and detection accuracy through multiscale learning, attention mechanisms, and contextual modeling. However, most of them treat SAR images as generic 2-D inputs, overlooking the underlying scattering mechanisms. More generally, the quality and fidelity of the input image can directly affect the reliability of downstream object detection; recent hyperspectral super-resolution work shows that material-proportion-guided unmixing can recover finer image structure and improve the usefulness of enhanced imagery for recognition tasks [35]. This observation is complementary to PASTE: the present work focuses on exploiting SAR scattering topology after image formation, while image-quality enhancement can be applied upstream when the acquisition or spatial resolution limits the separability of scattering responses.

State-space models (SSMs) have recently emerged as an efficient alternative for long-range representation learning [36], [37]. Mamba introduces selective SSMs with linear scaling in sequence length [36], while VMamba adapts selective scanning to 2-D visual features [38]. For object detection, Mamba-YOLO integrates SSM blocks into a YOLO-style detector [39], and OriMamba further targets oriented object detection in optical remote-sensing imagery [40]. These studies show the potential of state-space architectures for contextual modeling, but they focus on representation learning or detector design and do not explicitly use physically grounded SAR scattering topology as a prior for feature modulation or supervision. PASTE is complementary to these architectures:

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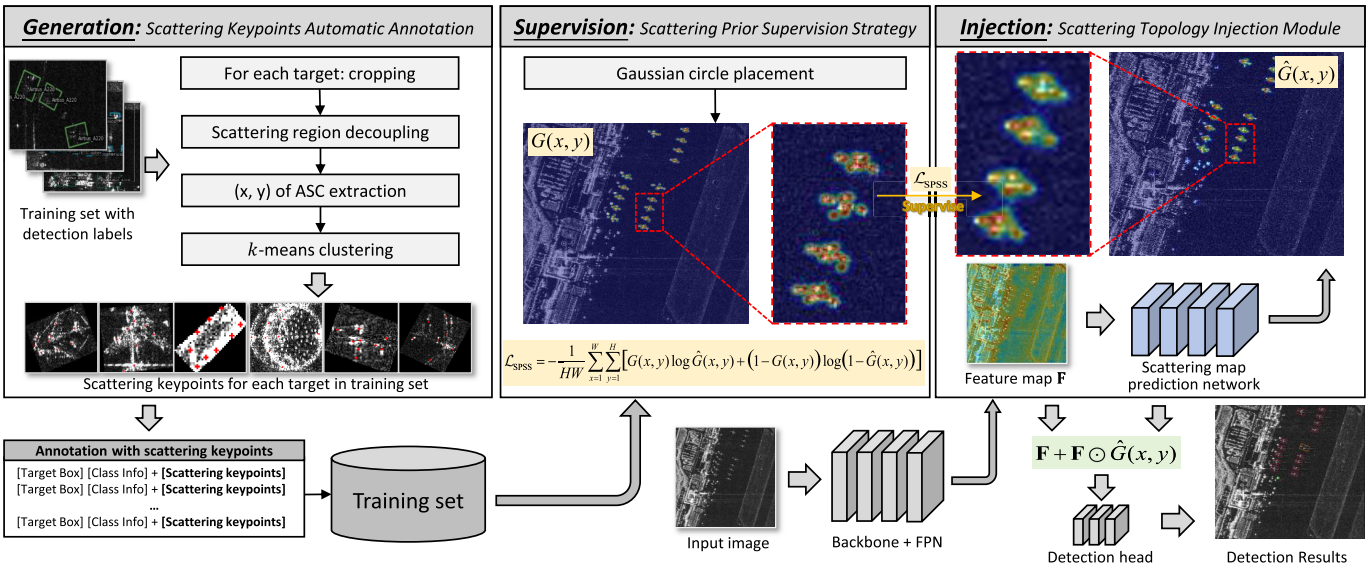


Fig. 2. Overview of the proposed PASTE framework. SKAA generates offline scattering topology priors from training annotations; STIM predicts and injects the corresponding scattering maps into FPN features; and SPSS converts the same keypoints into Gaussian targets to supervise the STIM map branch during training. (a) SKAA: each labeled target is cropped, decoupled, fit with ASCs, and clustered into stable keypoints. (b) SPSS: the keypoints form a continuous supervision map $\boldsymbol{G}(x, y)$. (c) STIM: the predicted map $\widehat{\boldsymbol{G}}(x, y)$ modulates the feature map $\boldsymbol{F}$ to enhance scattering-relevant regions.

it derives ASC-based topology from SAR targets and uses the resulting prior to guide detector features and optimization.

To better capture structural information, several works introduce keypoint-based or topology-aware representations. For example, Kang et al. [41] model relationships among scattering points, while Meng et al. [42] construct star-shaped topological structures. Xiao et al. [43] utilize reciprocal points to define structured feature spaces. Although these methods enhance structural modeling, their keypoints are typically derived from image-level statistics, making them heuristic and lacking explicit physical meaning.

Recent work by Ni et al. [44] introduces a size-aware graph embedding framework that incorporates object attribute information into remote sensing image captioning. While this approach emphasizes structured and attribute-aware representations, it focuses on high-level semantic modeling rather than integrating physically grounded scattering topology into detection frameworks. In contrast, our method explicitly models scattering topology based on ASC and embeds it into feature learning and supervision.

C. Methods Incorporating Scattering Characteristics

To incorporate physical priors, several studies explore scattering-based representations in SAR detection. Existing approaches can be broadly categorized into two types: 1) heuristic keypoint or scattering-enhanced methods and 2) ASC-based or physics-aware fusion methods.

Heuristic methods extract scattering regions or keypoints using image processing techniques. For example, Fu et al. [45] enhance scattering regions via Harris–Laplace detection, while Guo et al. [46] utilize scattering keypoints for anchor-free detection. These methods improve detection performance but rely on heuristic definitions of scattering structures.

In contrast, ASC-based methods aim to incorporate physically meaningful scattering representations. For instance, Wang et al. [47] use ASC masks for feature extraction, and Huang et al. [48] introduce physics-guided learning with scattering priors. Although these approaches incorporate physical information, they are typically limited to preprocessing or partial feature enhancement, lacking a unified integration into both feature learning and optimization.

In addition, recent work on physical constraint modeling, such as translation-difference characterization for PolSAR ship extraction [49], introduces scattering-related constraints for specific tasks. However, these methods are often task-specific and do not provide a general framework for integrating scattering topology into modern detectors.

In contrast to the above methods, this article proposes a unified ASC-based scattering topology embedding framework. PASTE connects electromagnetic scattering representations with detector feature learning through a pipeline of topology generation, feature injection, and prior-guided supervision.

III. Proposed Method

This section first presents an overview of PASTE and then details its three components: SKAA for scattering topology prior generation, STIM for multiscale feature injection, and SPSS for prior-guided supervision of the scattering-map branch.

A. Overall Architecture of PASTE

The overall architecture of PASTE is illustrated in Fig. 2. It consists of SKAA, the STIM, and the SPSS.

PASTE follows a concise prior-generation, feature-injection, and supervision workflow. First, SKAA converts target-level rotated-box annotations in a training split into ASC-derived

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scattering keypoints, producing the scattering topology prior offline. Next, STIM predicts spatial scattering maps from FPN features and injects them back into the corresponding feature levels. During training, SPSS turns the SKAA keypoints into Gaussian targets and supervises this map-prediction branch together with the detection losses. Thus, SKAA supplies the prior, STIM injects its learned representation, and SPSS closes the training loop by enforcing consistency between the predicted map and the prior.

B. Scattering Keypoints and Automatic Annotation Method

The complete SKAA workflow is illustrated in Fig. 3. The SKAA module automatically generates scattering keypoints from SAR targets and exports them as topology-aware supervision labels compatible with standard detection pipelines. The procedure consists of five steps: target-chip cropping, scattering-region decoupling, ASC position estimation, clustering to fixed keypoints, and label export. Residual updating is performed within the scattering-region decoupling step.

1) Step 1 (Crop Target Chip): Given the original SAR image and the corresponding bounding-box or rotated-box annotations, we first crop each target into an individual chip for scattering analysis. The crop region is slightly enlarged with a margin ratio to preserve the complete scattering extent of the target and to avoid truncating dominant scattering responses near the annotation boundary. In practice, the margin is set empirically according to the annotation format and image resolution, so that the cropped chip contains the full target and a limited amount of surrounding background.

For a rotated box parameterized by its center vector $\boldsymbol{c}$, width w, height h, and angle θ, let $\boldsymbol{u}_i$ be the local coordinate vector of an ASC in the padded chip. With horizontal and vertical margins m_x and m_y, the chip dimensions are $W_c = w + 2m_x$ and $H_c = h + 2m_y$. The ASC coordinate is mapped back to the original image by

$$\begin{aligned} \boldsymbol{p}_i^{\text{img}} &= \boldsymbol{c} + \boldsymbol{Q}(\theta)\left(\boldsymbol{u}_i - \tfrac{1}{2}\boldsymbol{d}_c\right) \\ \boldsymbol{Q}(\theta) &= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \\ \boldsymbol{d}_c &= [W_c, H_c]^{\mathsf{T}} \end{aligned} \tag{1}$$

where the rotation convention is the same as that used by the source oriented-box annotation. Any image-space augmentation or resize applied by the detector is applied identically to the keypoint coordinates, denoted by $\boldsymbol{p}_i^{\text{net}} = \mathcal{A}(\boldsymbol{p}_i^{\text{img}})$. For an FPN level with stride s_l, the coordinate used to build the supervision map is $\boldsymbol{p}_i^{(l)} = \boldsymbol{p}_i^{\text{net}}/s_l$. This explicit chain preserves the correspondence from the rotated target chip to the original image and then to each feature map.

This chip-level operation provides two benefits. First, it reduces the interference from neighboring targets and large-scale clutter. Second, it makes the subsequent ASC fitting more stable, since the optimization is restricted to a localized region centered on a single target. Let the cropped target chip be denoted as $\boldsymbol{I} \in \mathbb{C}^{H\times W}$.

2) Step 2 (Scattering-Region Decoupling): After cropping the target chip, we perform scattering-region decoupling to isolate dominant local scattering structures from background clutter and weak residual responses. This procedure is implemented by the image-domain decoupling algorithm summarized in Algorithm 1.

Algorithm 1 Image-Domain Scattering Region Decoupling

Input: Complex image $\boldsymbol{I} \in \mathbb{C}^{H\times W}$, dB threshold τ
Output: Decoupled scattering region list $\mathcal{S}$
$\boldsymbol{R} \leftarrow |\boldsymbol{I}|$, $\mathcal{S} \leftarrow \emptyset$, $n \leftarrow 0$;
while $n < N_{\max}$ **do**
  (1) Scattering Block Masking based on Breadth First Search (BFS);
  $\tau_p \leftarrow \max(\boldsymbol{R}) \cdot 10^{\tau/10}$;
  $(x_0, y_0) \leftarrow \arg\max \boldsymbol{R}$;
  Initialize binary mask $\boldsymbol{B} \leftarrow 0$;
  Initialize queue $\mathcal{Q} \leftarrow \{(x_0, y_0)\}$, set $\boldsymbol{B}(x_0, y_0) = 1$;
  while $\mathcal{Q}$ *not empty* **do**
    Pop (x, y) from $\mathcal{Q}$;
    foreach 4-*neighbor* (x', y') *of* (x, y) **do**
      if (x', y') *out of bounds* **then** continue
      if $\boldsymbol{B}(x', y') = 1$ **then** continue
      if $\boldsymbol{R}(x', y') \le \tau_p$ **then** continue
      $\boldsymbol{B}(x', y') \leftarrow 1$;
      Push (x', y') into $\mathcal{Q}$;
  (2) Log-amplitude guided region growing;
  $p \leftarrow \max(\boldsymbol{R})$;
  $\boldsymbol{D}(x, y) \leftarrow 10\log_{10}\big((\boldsymbol{R}(x, y) + \epsilon)/p\big)$;
  $\Omega \leftarrow \{(x, y) \mid \boldsymbol{D}(x, y) > -20\}$ (sorted desc.);
  $\boldsymbol{T} \leftarrow \boldsymbol{B}$, $l \leftarrow 1$;
  foreach $(x, y) \in \Omega$ **do**
    if $\boldsymbol{T}(x, y) \neq 0$ **then** continue
    $\mathcal{N} \leftarrow$ labels of 8-neighbors of (x, y), remove 0;
    if $\mathcal{N} = \emptyset$ ***and*** $\boldsymbol{D}(x, y) > \tau$ **then**
      $l \leftarrow l + 1$, $\boldsymbol{T}(x, y) \leftarrow l$
    else
      $\boldsymbol{T}(x, y) \leftarrow \min(\mathcal{N})$
  (3) Region extraction and residual update;
  $\boldsymbol{S}(x, y) \leftarrow \boldsymbol{R}(x, y)$ if $\boldsymbol{T}(x, y) = 1$, else 0;
  $\mathcal{S} \leftarrow \mathcal{S} \cup \{\boldsymbol{S}\}$;
  $\boldsymbol{R} \leftarrow \max(\boldsymbol{R} - \boldsymbol{S}, 0)$;
  $n \leftarrow n + 1$;
return $\mathcal{S}$
Parameter setting: $\tau = -3$ dB, $\epsilon = 10^{-6}$, $N_{\max} = 20$.

Given the cropped complex-valued target chip $\boldsymbol{I} \in \mathbb{C}^{H\times W}$, we first compute its amplitude image $\boldsymbol{R} = |\boldsymbol{I}|$ and iteratively extract dominant scattering regions from $\boldsymbol{R}$. In each iteration, the current strongest scattering response is selected as the seed, and a binary mask is initialized from the peak position. A breadth-first search (BFS) is then performed to identify the connected high-response block around this peak. Specifically,

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the local masking threshold is defined as

$$\tau_p = \max(\boldsymbol{R}) \cdot 10^{\tau/10}$$

where τ is the decoupling threshold in dB. Starting from the peak position, neighboring pixels are added in the mask if their amplitudes exceed τ_p, which yields a compact initial scattering block associated with the dominant local scatterer.

To further recover the complete extent of the scattering region, we then perform log-amplitude guided region growing. Let $p = \max(\boldsymbol{R})$ denote the current peak response, and define the normalized log-amplitude map as

$$\boldsymbol{D}(x, y) = 10 \log_{10}((\boldsymbol{R}(x, y) + \epsilon)/p)$$

where ϵ is a small constant introduced for numerical stability. Pixels with sufficiently high relative responses are collected and sorted in descending order. Based on the BFS mask as the initial seed, connected pixels are progressively assigned to the target region according to their neighboring labels, which allows the dominant scattering structure to expand adaptively while suppressing isolated noise responses.

After region growing, the extracted dominant scattering region $\boldsymbol{S}$ is recorded, and the residual amplitude image is updated by subtracting the extracted region from the current residual

$$\boldsymbol{R} \leftarrow \max(\boldsymbol{R} - \boldsymbol{S}, 0).$$

This residual update is crucial because it prevents previously extracted strong scatterers from repeatedly dominating subsequent iterations and helps reveal secondary but still meaningful scattering structures. The above process is repeated until the maximum iteration number $N_{\max}$ is reached or no significant scattering response remains in the residual image.

In practice, the decoupling process is controlled by three key factors: the dB threshold τ, the residual update rule, and the stopping condition. The threshold τ determines the spatial extent of the extracted scattering block. A larger threshold produces a more compact but conservative region, whereas a smaller threshold preserves more weak scattering responses at the cost of increased noise sensitivity. The residual update suppresses already-explained dominant peaks and improves the diversity of extracted regions. The stopping condition is implemented through the iteration cap $N_{\max}$ together with the progressive attenuation of the residual response. Unless otherwise specified, we set $\tau = -3$ dB, $\epsilon = 10^{-6}$, and $N_{\max} = 20$ following Algorithm 1. These values are fixed across categories rather than tuned per class: $\tau = -3$ dB retains connected dominant responses while rejecting weaker background fluctuations, and $N_{\max} = 20$ is a conservative cap that prevents excessive decomposition after the residual has become weak. With $N_{\max} = 20$ and $k = 9$ fixed, $\tau = -1$, -3, and -5 dB yield 42.26%, 43.30%, and 42.53% mean average precision (mAP)@0.5, respectively; with $\tau = -3$ dB and $k = 9$ fixed, $N_{\max} = 10$, 20, and 30 yield 42.84%, 43.30%, and 42.58%, respectively.

3) Step 3 (ASC Position Estimation): Given the decoupled scattering region list $\mathcal{S} = \{\boldsymbol{S}_i\}$ obtained in Step 2, we estimate the spatial positions of dominant ASCs by fitting each region with a parametric scattering model. Instead of directly performing global optimization on the entire target chip, the estimation is carried out locally on each decoupled scattering region, which significantly improves stability and computational efficiency.

For each scattering region $\boldsymbol{S}_i$, the ASC position $\boldsymbol{p}_i = (x_i, y_i)^{\mathsf{T}}$ is estimated by minimizing the reconstruction error between the observed scattering feature and the model-generated response

$$\boldsymbol{p}_i = \arg\min_{(x,y)} \|\boldsymbol{S}_i - \widehat{\boldsymbol{S}}_i(x, y)\|_2 \tag{2}$$

where $\widehat{\boldsymbol{S}}_i(x, y)$ denotes the scattering response synthesized from the ASC model at location (x, y). Specifically, the ASC response in the frequency domain is modeled as

$$\boldsymbol{E}_i(x_i, y_i) = C(f, \varphi) \cdot \exp\left(-\mathrm{j}\frac{4\pi f}{\mathrm{c}}(x_i \cos\varphi + y_i \sin\varphi)\right) \tag{3}$$

and the corresponding image-domain scattering feature is obtained via

$$\widehat{\boldsymbol{S}}_i(x_i, y_i) = \mathrm{IFFT2D}(w(\boldsymbol{E}_i(x_i, y_i))) \tag{4}$$

where $w(\cdot)$ denotes the applied window function.

In practice, (2) is solved using a localized search strategy rather than exhaustive global optimization. The initialization of (x_i, y_i) is obtained from the peak location of the corresponding decoupled region S_i, which provides a physically meaningful starting point. Around this initialization, a small spatial neighborhood is explored to refine the ASC position by evaluating the reconstruction error in (2). Specifically, the single-scattering-center iterative optimization algorithm proposed in [50] is adopted.

The objective in (2) estimates stable ASC positions that represent the dominant scattering topology of the target; it is not intended to reconstruct every scattering detail. Region-level decoupling and localized optimization reduce the influence of noise and minor fluctuations. The estimated (x, y) coordinates are then passed to the clustering stage.

4) Step 4 (Clustering to Fixed Keypoints): The iterative estimation procedure may produce a variable number of ASC candidates for different targets. Although an adaptive number of keypoints could be considered, a fixed setting is adopted to ensure consistent supervision and stable training across targets with varying scattering complexity. Let $\mathcal{Z} = \{\boldsymbol{z}_i\}_{i=1}^{n}$ be the ASC position vectors obtained in Step 3. We obtain k representative keypoints by minimizing the standard within-cluster spatial dispersion

$$\{\boldsymbol{\mu}_j\}_{j=1}^{k} = \arg\min_{\{\boldsymbol{\mu}_j\}_{j=1}^{k}} \sum_{i=1}^{n} \min_{j \in \{1,\ldots,k\}} \|\boldsymbol{z}_i - \boldsymbol{\mu}_j\|_2^2 \tag{5}$$

where the cluster centers $\boldsymbol{\mu}_j$ are the final scattering keypoints. We use k-means++ initialization and retain the minimum-dispersion solution across repeated initializations. The resulting keypoints inherit their physical basis from the ASC candidates, while clustering provides a

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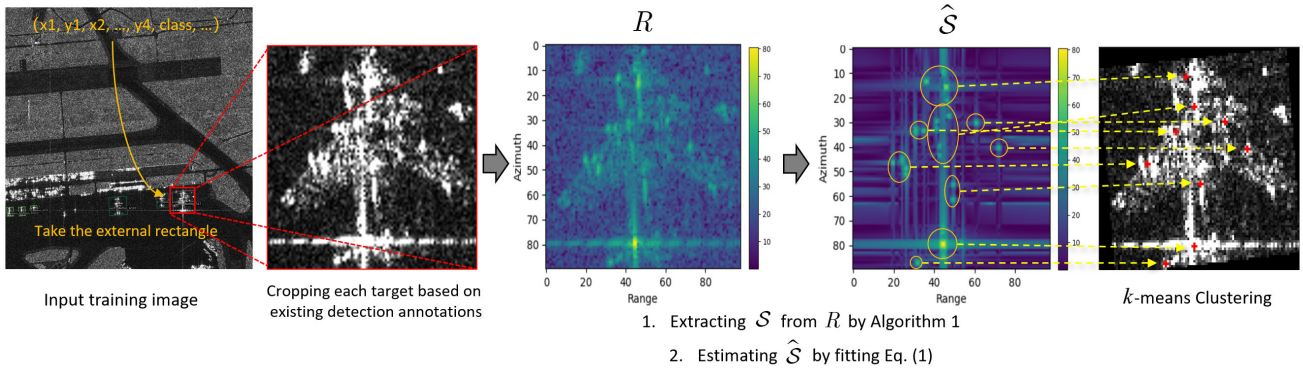


Fig. 3. Workflow of the SKAA method.

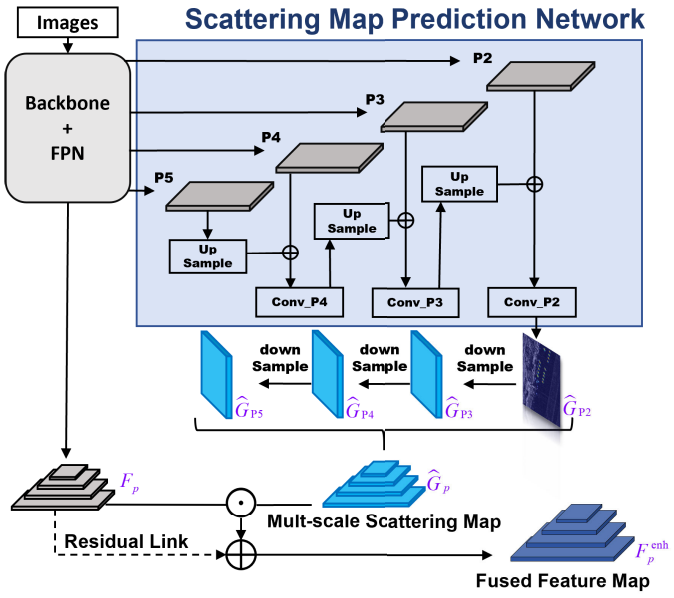


Fig. 4. Scattering map prediction network. Input: multiscale features $\boldsymbol{F}_p$ from the backbone with FPN. A top-down pathway with upsampling and lateral fusion progressively propagates high-level semantics from $P5$ to $P2$, followed by convolution layers at each level to predict scattering maps $\widehat{\boldsymbol{G}}_p$. Output: multiscale scattering maps $\widehat{\boldsymbol{G}}_p$, which are further used to enhance features via element-wise modulation, yielding refined features $\boldsymbol{F}_p^{\text{enh}}$ for subsequent detection.

consistent fixed-cardinality representation of their spatial topology.

The use of a fixed number of keypoints has three purposes. First, it provides a unified topology representation that simplifies label construction and batch training. Second, it prevents target-dependent variations in the number of estimated scatterers from causing inconsistent supervision. Third, it retains the dominant spatial structure while suppressing redundant or unstable candidates. The choice of k is determined empirically from the trade-off between structural completeness and annotation stability. A small k may miss important scattering structure, whereas a large k may introduce redundancy and amplify noise sensitivity. The ablation in Section IV identifies $k = 9$ as the best setting among those evaluated.

5) Step 5 (Label Export): Finally, the clustered keypoints are exported as extended annotations for detector training. Specifically, we retain the original DOTA-style [51] instance annotation format and append the generated scattering keypoints to the end of each annotation line. Therefore, each line still corresponds to one target instance, while the annotation is augmented with k additional keypoint coordinate pairs $(x_1, y_1), (x_2, y_2), \ldots, (x_k, y_k)$ after the original object fields. The conventional detection labels and the generated scattering-topology priors are therefore stored in a unified instance-level format. A lightweight parser extension makes the format compatible with the training pipeline.

For a target originally annotated in the DOTA format as

$$x_1^b \; y_1^b \; x_2^b \; y_2^b \; x_3^b \; y_3^b \; x_4^b \; y_4^b \; c \; d$$

where (x_j^b, y_j^b) denotes the four box vertices, c is the category label, and d is the difficulty flag, the extended annotation is written as

$$x_1^b \; y_1^b \; x_2^b \; y_2^b \; x_3^b \; y_3^b \; x_4^b \; y_4^b \; c \; d \; x_1 \; y_1 \; x_2 \; y_2 \; \cdots \; x_k \; y_k.$$

SKAA requires no manual keypoint annotation. The exported labels serve as topology priors for STIM and are converted into soft supervision targets for SPSS, linking ASC-based modeling to detector training.

SKAA is an offline label-generation step. It is run once for the training split of a new dataset, an updated annotation set, or a new imaging configuration that requires adaptation. The resulting extended annotations can then be reused for all training runs and detector variants on that prepared split. SKAA is not run on test images or on incoming images at deployment; both ASC fitting and clustering are absent from the inference path.

C. Scattering Topology Injection Module

STIM incorporates the scattering topology prior into the feature representation by predicting a spatial scattering map and using it to modulate multiscale feature maps. The modulation reinforces responses near the predicted scattering structures.

STIM operates on intermediate feature maps and leaves the backbone and detection head unchanged. It receives multiscale FPN features, predicts the scattering map, and returns enhanced features at the corresponding levels.

The detailed architecture of the scattering map prediction network in STIM and its fusion flow is illustrated in Fig. 4.

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For clarity, we denote the multiscale FPN feature at level p as $\boldsymbol{F}_p$ and the predicted scattering map as $\widehat{\boldsymbol{G}}_p$.

The design rationale of this network is as follows. Shallow feature maps contain richer spatial details, making them suitable for predicting the spatial distribution of target scattering areas. This is particularly important for SAR imagery, where scattering responses are spatially sparse and highly localized. Predicting the scattering map at shallow layers ensures that fine-grained spatial structures of scattering centers are preserved. Meanwhile, high-level features carry stronger semantic information that provides complementary contextual guidance.

Therefore, a progressive upsampling and concatenation strategy is adopted to integrate semantic cues from deep layers into the shallow ones, enabling the scattering map to jointly capture both spatial and semantic characteristics. Specifically, high-level features (e.g., P5) are progressively upsampled and fused with lower-level FPN features (P4, P3, P2). Each fusion step is followed by a 3×3 convolution (Conv_P4, Conv_P3, Conv_P2) and ReLU activation. Finally, a 1×1 convolution and sigmoid function generate the spatial scattering map at the shallowest level.

To achieve efficient multiscale scattering map generation, the scattering map is predicted only at the shallowest scale, where spatial details are most abundant. This design avoids redundant computation while maintaining spatial fidelity, since higher-level maps can be effectively derived through progressive downsampling without introducing additional learnable parameters. As shown in Fig. 4, scattering maps for higher levels are obtained via downsampling, thus forming a spatial attention pyramid consistent with the FPN hierarchy.

Afterward, the predicted spatial scattering map is applied to the original FPN features via element-wise multiplication to emphasize scattering-relevant regions. The enhanced features are then added back to the original FPN outputs, forming the final enhanced feature $\boldsymbol{F}_p^{\text{enh}}$. The process can be expressed as follows:

$$\boldsymbol{F}_p^{\text{enh}} = \boldsymbol{F}_p + \widehat{\boldsymbol{G}}_p \odot \boldsymbol{F}_p, \quad p \in \{P2, P3, P4, P5\}. \tag{6}$$

This residual formulation preserves the original feature representation while selectively enhancing scattering-relevant regions.

D. Scattering Prior Supervision Strategy

During inference, neither SKAA nor SPSS is executed. The scattering topology enhancement is produced solely by the trained STIM map-prediction branch, which has learned the scattering topology prior from the offline annotations during training. Thus, PASTE retains an end-to-end inference path without ASC fitting, clustering, or external keypoint labels for each input image.

To enable the network to effectively capture the distribution of scattering keypoints while maintaining compatibility with downstream detection objectives, a joint optimization strategy is adopted. Specifically, the proposed SPSS is combined with the detection loss to supervise the training of the scattering map prediction branch.

During training, SPSS constructs a ground-truth (GT) scattering map to guide the predicted scattering map $\widehat{\boldsymbol{G}}_p$ toward the scattering keypoint locations. Ideally, the predicted attention distribution should exhibit high responses only around the scattering keypoints generated by SKAA. Thus, SPSS initializes a zero-valued map $\boldsymbol{G}_p$ with the same spatial size as the feature map.

Let the feature-map size be $H \times W$, and define the scattering-keypoint set $\mathcal{K}$ as

$$\mathcal{K} = \{\boldsymbol{p}_k\}_{k=1}^{N}. \tag{7}$$

Let $\boldsymbol{q} = [x, y]^{\mathsf{T}}$ denote a feature-map coordinate vector. Then the GT Gaussian scattering map $\boldsymbol{G}_p(x, y)$ is defined as

$$\boldsymbol{G}_p(x, y) = \max_{k=1,\ldots,N} \exp\left(-\frac{\|\boldsymbol{q} - \boldsymbol{p}_k\|_2^2}{2\sigma^2}\right), \quad x \in [0, W),\ y \in [0, H) \tag{8}$$

where W and H denote the spatial dimensions of the feature map, and σ controls the spatial spread of each scattering keypoint. Note that σ is defined on the feature map rather than in the input image. For an FPN level with stride s_l, one feature-map pixel corresponds to s_l pixels in the input image, so the Gaussian standard deviation in the image space is $s_l\sigma$. Accordingly, for a response level $\gamma \in (0, 1)$, the corresponding radius in the input image is $r_\gamma^{\text{img}} = s_l\sigma(-2\ln\gamma)^{1/2}$. In particular, on the P2 feature level with $s_{P2} = 4$, this becomes $r_\gamma^{\text{img}} = 4\sigma(-2\ln\gamma)^{1/2}$. Therefore, when $\sigma = 1$, the Gaussian standard deviation corresponds to 4 pixels in the input image. In practice, we use a relatively small σ because even the shallowest FPN feature map is already downsampled, and a larger spread would reduce the localization precision of scattering keypoints. Accordingly, σ is initialized as 1, and its final value is determined through the ablation study in Section IV.

The resulting GT scattering map $\boldsymbol{G}_p$ provides a smooth and continuous supervision signal that captures the spatial topology of scattering keypoints. The learning objective of the scattering map prediction branch is defined using binary cross-entropy (BCE) loss

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{HW}\sum_{x=1}^{W}\sum_{y=1}^{H}\Big[\boldsymbol{G}_p(x, y)\log\widehat{\boldsymbol{G}}_p(x, y) + \big(1 - \boldsymbol{G}_p(x, y)\big)\log\Big(1 - \widehat{\boldsymbol{G}}_p(x, y)\Big)\Big]. \tag{9}$$

During training, the scattering-map loss is applied at the shallowest FPN level. The input features to this auxiliary branch are detached, so this loss updates the map-prediction branch without directly updating the backbone or FPN through the auxiliary loss.

IV. Experimental Results

This section evaluates PASTE through quantitative comparisons, qualitative analyses, ablation studies, robustness tests, and resource measurements. We first introduce the dataset and training procedure, including visualizations of the scattering keypoint annotations generated by SKAA. We then define the evaluation metrics and compare the selected baseline detectors with and without PASTE. Finally, we analyze the predicted spatial scattering maps, component and parameter ablations,

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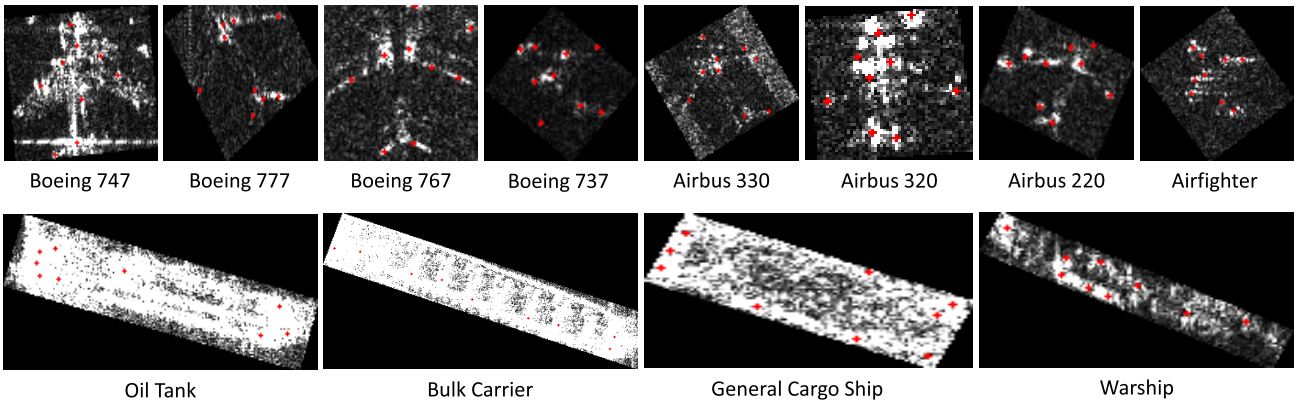


Fig. 5. Visualization of scattering keypoints for representative classes in the FAIR-CSAR training set. The first row shows examples from major aircraft categories, while the second row illustrates keypoint patterns of representative ship categories. All automatically generated scattering keypoints are marked with red dots in the figure.

robustness to ASC localization error, and computational overhead.

A. Datasets

The FAIR-CSAR [52] dataset, a fine-grained object detection and recognition benchmark built from single-look complex (SLC) SAR imagery, is used in our experiments. It addresses several limitations of existing SAR datasets, including the absence of signal-domain information, coarse annotation granularity, and insufficient sample volume. FAIR-CSAR is constructed from 175 SLC images acquired by the Gaofen-3 satellite, covering 32 cities worldwide as well as multiple maritime regions. The dataset contains 349 002 annotated instances with spatial resolutions ranging from 1 to 5 m and employs oriented bounding boxes (OBBs) to support object analysis in complex scenes.

FAIR-CSAR consists of two imaging modes of Gaofen-3: SL and Fine Strip-Map I (FSI).

1) SL Mode: The SL mode is the primary high-resolution acquisition protocol of FAIR-CSAR and is designed for fine-grained object analysis in complex scenes. It provides a nominal spatial resolution of 1 m and uses single-polarization acquisitions covering four polarimetric combinations (HH, HV, VH, and VV). The current release contains approximately 170k annotated instances distributed across 22 fine-grained categories, namely Airbus_A220, Airbus_A320, Airbus_A330, Airfreighter, Boeing_737, Boeing_747, Boeing_767, Boeing_777, Fokker-50, Gulfstream, Helicopter, Other_Aircraft, Bulk_Carrier, Container_Ship, General_Cargo_Ship, Motion-Defocusing_Ship, Oil_Tanker, Warship, Other_Ship, Bridge, Tank, and Tower_Crane.

2) FSI Mode: The FSI mode is tailored for large-area surveillance at medium resolution. Acquired with a nominal ground-range resolution of 5 m and dual-polarimetric channels, this mode contains 170k labeled instances condensed into three major categories: Bridge, Ship, and Tank. FSI imagery spans oceans, rivers, and harbors, and faithfully reproduces real-world SAR phenomena such as motion defocusing and azimuth ambiguities.

3) Complementarity of SL and FSI: The SL and FSI modes use distinct SAR imaging mechanisms. The SL mode steers the radar beam to continuously illuminate a fixed scene, enabling a long coherent integration time and thus high spatial resolution. In contrast, the FSI mode scans along the flight path with a wider swath, resulting in a shorter integration time, lower resolution, and more pronounced imaging artifacts.

These modes provide complementary evaluation conditions: SL emphasizes detailed representation learning, whereas FSI tests robustness under lower resolution and stronger imaging artifacts. Together, they cover two resolutions and polarization settings as well as different scene complexities and category granularities within FAIR-CSAR.

The two imaging modes provide complementary conditions, but they do not cover all possible sensors or acquisition settings. When scattering centers are not spatially resolved, adjacent responses may merge and reduce the usefulness of sparse topology priors. Applying PASTE to an unseen imaging configuration therefore requires generating SKAA annotations for the corresponding training split and retraining or fine-tuning the detector.

B. Training Details

The six CNN- and Transformer-based detector pairs are trained for 15 epochs on the FAIR-CSAR SL and FSI training sets, and the checkpoint achieving the best performance during training is selected for final evaluation. The batch size is fixed to 4 for all configurations. For detectors based on ResNet-50, we use an stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.005, momentum of 0.9, and weight decay of 1×10^{-4}, together with gradient clipping ($\text{norm}_{\max} = 35$). The learning rate follows a step decay schedule with linear warm-up for the first 500 iterations and decay points at epochs 8 and 11. For Swin-T–based detectors, AdamW is adopted with a learning rate of 1×10^{-4}, weight decay of 0.05, and customized parameter-wise decay settings for positional embeddings and normalization layers. All experiments are conducted on a Linux server equipped with two NVIDIA RTX 2080 Ti GPUs.

We also adapt the official Mamba-YOLO implementation [39] to FAIR-CSAR OBBs using the Ultralytics OBB head. Both standalone Mamba-YOLO-M and its PASTE-

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TABLE I
BASELINE DETECTORS ADOPTED FOR PASTE EVALUATION

Baseline Detector	Stage	Anchor	Justification for Selection
Rotated RetinaNet	single	anchor-based	Classic one-stage anchor-based baseline; widely used in rotated object detection.
GWD	single	anchor-free	Represents anchor-free regression with probabilistic localization loss.
RoI Transformer	two	anchor-based	Converts horizontal RoIs to rotated ones via learnable transformation; tests PASTE on geometry-aware proposal refinement.
Rotated FCOS	single	anchor-free	Dense anchor-free detector; tests PASTE on center-ness assignment.
Oriented R-CNN (ResNet-50)	two	anchor-based	Strong two-stage CNN baseline; demonstrates PASTE compatibility with standard backbones.
Oriented R-CNN (Swin-T)	two	anchor-based	Swin-Transformer variant; evaluates PASTE on Transformer backbones.
Mamba-YOLO-M	single	anchor-free	Recent state-space-model detector; its standalone and PASTE-enhanced OBB variants test compatibility with state-space backbones.

TABLE II
BASELINE COMPARISON AND PERFORMANCE BOOST BY PASTE ON FAIR-CSAR (SL MODE)

Baseline Detector	mAP@0.5	mAP@0.5 (+PASTE)	Gain	Relative gain (%)
Rotated RetinaNet	22.9	24.0	**1.1** ↑	4.8%
GWD	26.0	27.0	**1.0** ↑	3.8%
RoI Transformer	38.0	42.3	**4.3** ↑	11.3%
Rotated FCOS	34.5	35.7	**1.2** ↑	3.5%
Oriented R-CNN (ResNet-50)	39.2	43.3	**4.1** ↑	10.5%
Oriented R-CNN (Swin-T)	41.3	44.4	**3.1** ↑	7.5%
Mamba-YOLO-M	39.6	40.1	**0.5** ↑	1.3%

TABLE III
BASELINE COMPARISON AND PERFORMANCE BOOST BY PASTE ON FAIR-CSAR (FSI MODE)

Baseline Detector	mAP@0.5	mAP@0.5 (+PASTE)	Gain	Relative gain (%)
Rotated RetinaNet	37.2	38.9	**1.7** ↑	4.6%
GWD	39.4	40.6	**1.2** ↑	3.0%
RoI Transformer	45.4	48.7	**3.3** ↑	7.3%
Rotated FCOS	45.4	46.7	**1.3** ↑	2.9%
Oriented R-CNN (ResNet-50)	44.8	48.3	**3.5** ↑	7.8%
Oriented R-CNN (Swin-T)	50.0	52.0	**2.0** ↑	4.0%
Mamba-YOLO-M	58.1	59.5	**1.4** ↑	2.4%

enhanced counterpart are initialized from scratch because the official repository does not provide a compatible OBB checkpoint. They use 1024×1024 inputs, a batch size of 4, seed 0, and SGD with an initial learning rate of 0.005, momentum of 0.9, and weight decay of 1×10^{-4}. To retain SAR intensity statistics, mosaic and color augmentation are disabled, while horizontal and vertical flips are retained. The PASTE-enhanced model retains the Mamba-YOLO-M backbone and OBB task and incorporates the ASC-derived topology through the enhanced OBB head. Both variants use the same FAIR-CSAR data splits and validation procedure.

Before training detectors equipped with the proposed PASTE, the training split of each dataset is first preprocessed by SKAA to automatically generate scattering keypoint annotations. This procedure is fully automated and produces all required supervision signals before training begins. Fig. 5 presents the visualized scattering keypoints generated by SKAA for selected FAIR-CSAR training categories.

C. Evaluation Metrics and Computation

Model performance is evaluated using the mAP, the standard metric for object detection. For each class c, the average precision (AP) is computed as the area under the precision–recall curve

$$\mathrm{AP}_c = \int_0^1 p_c(r)\, \mathrm{d}r \tag{10}$$

where $p_c(r)$ denotes the precision at recall level r. The mAP is obtained by averaging (10) over C classes

$$\mathrm{mAP} = \frac{1}{C} \sum_{c=1}^{C} \mathrm{AP}_c. \tag{11}$$

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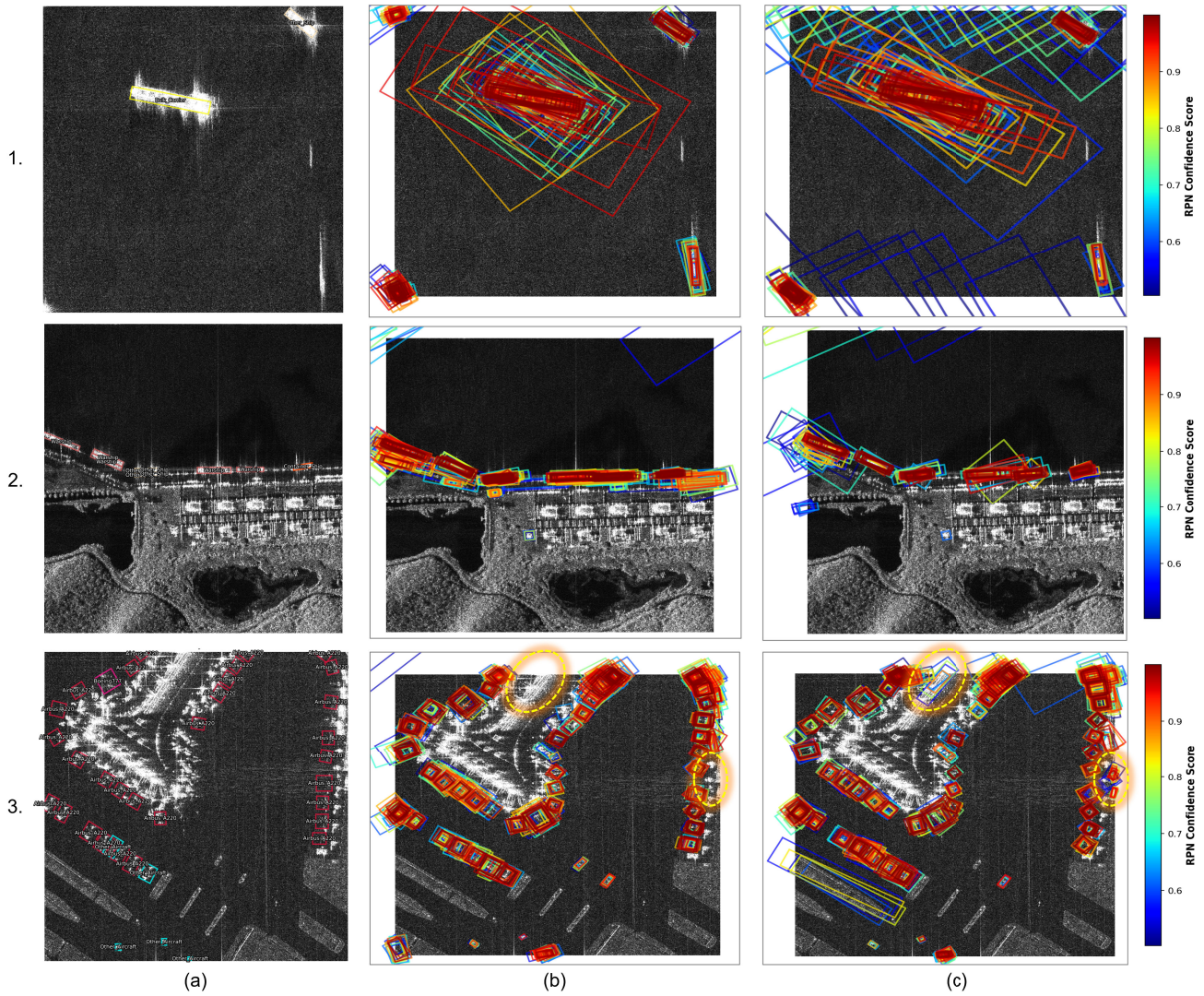


Fig. 6. Visualization of first-stage proposals in three scenes. Row 1: inshore ships. Row 2: offshore ships. Row 3: terminal aircraft. (a) GT boxes. (b) Top-scoring proposals (≥ 0.5) generated by Oriented R-CNN equipped with PASTE. (c) Top-scoring proposals (≥ 0.5) generated by Oriented R-CNN without PASTE. All proposals are colored from cool to warm according to confidence score.

In our experiments, we adopt mAP@0.5, where a predicted oriented bounding box $\widehat{\boldsymbol{B}}$ is considered correct if its intersection-over-union (IoU) with the GT box $\boldsymbol{B}$ satisfies

$$\mathrm{IoU}\left(\boldsymbol{B}, \widehat{\boldsymbol{B}}\right) = \frac{|\boldsymbol{B} \cap \widehat{\boldsymbol{B}}|}{|\boldsymbol{B} \cup \widehat{\boldsymbol{B}}|} \geq 0.5. \tag{12}$$

This metric jointly captures both localization accuracy and classification performance.

D. Baseline Methods and Comparative Experiments

This section evaluates the effectiveness and architectural compatibility of PASTE. We select baseline detectors that span single-stage and two-stage paradigms, CNN-based and Transformer-based backbones, SSMs, and anchor-based and anchor-free designs. The selected detectors and their main characteristics are summarized in Table I, along with the rationale for their selection.

All baseline detectors in Table I were trained following the configurations described in Section IV-B. The six CNN- and Transformer-based detectors were trained again with PASTE under the same conditions, while Mamba-YOLO-M was evaluated in standalone and PASTE-enhanced forms with the same backbone, OBB task, data splits, and validation procedure. The resulting SL and FSI comparisons are reported in Tables II and III, respectively.

Tables II and III show that PASTE improves mAP@0.5 for every evaluated detector pair under both FAIR-CSAR modes. On SL, the absolute gains range from 0.5 to 4.3 percentage points (1.3%–11.3% relative improvement). RoI Transformer obtains the largest increase, from 38.0% to 42.3% (+4.3 percentage points; 11.3% relative), followed by Oriented R-CNN with ResNet-50, from 39.2% to 43.3% (+4.1 percentage points; 10.5% relative). The additional state-space-model pair also follows this trend, improving from 39.6% to 40.1%.

The same pattern is observed on FSI, where the absolute improvements range from 1.2 to 3.5 percentage points and the relative improvements from 2.4% to 7.8%. The largest FSI gain is obtained by Oriented R-CNN with ResNet-50

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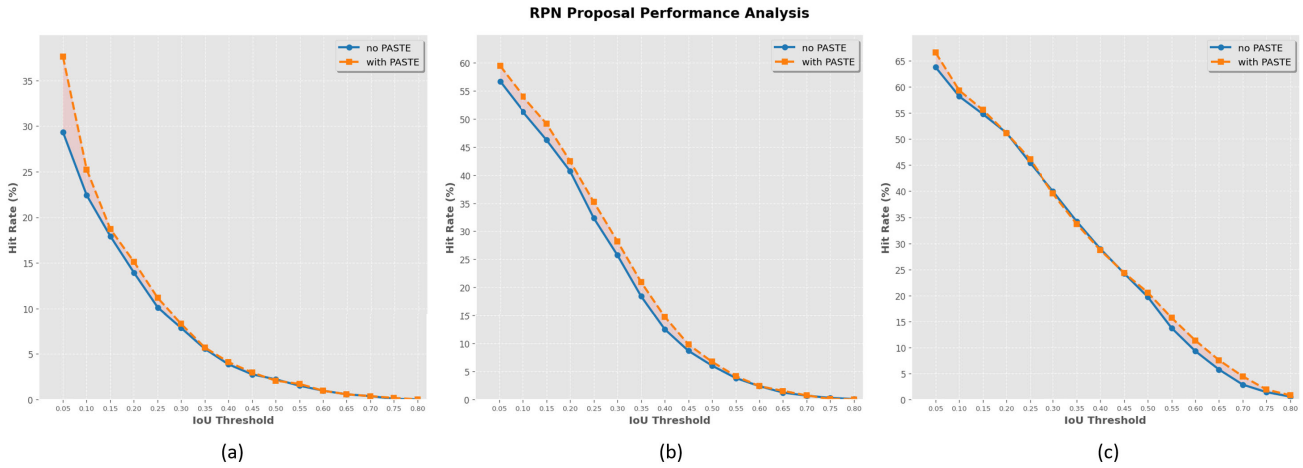


Fig. 7. PHR under varying IoU thresholds in three representative scenes (a) offshore ships, (b) inshore ships, and (c) terminal aircraft. The baseline detector is Oriented R-CNN (ResNet-50).

(+3.5 percentage points; 7.8% relative), followed by RoI Transformer (+3.3 percentage points; 7.3% relative). The state-space-model pair increases from 58.1% to 59.5%. Taken together, these paired comparisons indicate that the proposed topology supervision is compatible with the evaluated one-stage and two-stage detectors, CNN, Transformer, and state-space backbones, as well as anchor-based and anchor-free designs.

The two-stage detectors show larger gains than the single-stage detectors in the evaluated pairs: 3.1–4.3 versus 1.0–1.2 percentage points on SL, and 2.0–3.5 versus 1.2–1.7 percentage points on FSI. We therefore analyze proposal generation using Oriented R-CNN checkpoints trained with and without PASTE on the same representative scenes.

1) Qualitative Analysis of Impact on Proposal Generation by PASTE: To examine proposal generation, we visualize the region proposal network (RPN) proposals together with the corresponding GT boxes in Fig. 6. Three representative scenes are shown: inshore ships, offshore ships, and terminal aircraft, arranged from top to bottom. From left to right, the columns show the GT boxes, proposals generated with PASTE, and proposals generated without PASTE.

Examining the first two rows (ship scenes), proposals generated with PASTE exhibit a high degree of shape consistency: their orientations and aspect ratios closely match and tightly cluster around the GT boxes. In contrast, proposals produced without PASTE are markedly more scattered, with divergent orientations and aspect ratios even at the same location. This indicates that PASTE injects scattering-topology geometric cues that impose an effective geometric constraint on the first-stage RPN.

In the terminal scene, the baseline RPN produces two high-confidence proposals in the strongly scattering region of the terminal building, whereas the PASTE-enhanced RPN does not produce proposals there. The PASTE-enhanced RPN also produces more proposals around small aircraft in the apron area. These examples illustrate how the scattering-topology prior changes proposal placement in cluttered regions.

2) Quantitative Analysis of Impact on Proposal Generation by PASTE: Following the qualitative visualization analysis, we quantify the effect of PASTE on RPN proposals across the three representative scenarios. We calculate the proposal hit rate (PHR) between all generated proposals and the GT boxes across IoU thresholds from 0.05 to 0.8, producing the curve in Fig. 7. PHR is the ratio of proposals whose IoU with any GT box exceeds the threshold to the total number of proposals. In the ship scenes, PASTE produces a higher hit rate for medium- and low-quality proposals with IoU below 0.5, while the hit rate for high-quality proposals remains comparable to the baseline. In the airport terminal scene, PASTE produces a higher hit rate below 0.15 and above 0.5 IoU, and comparable performance at the remaining thresholds. These results suggest that PASTE improves the discriminative capability of the detector when proposal quality is limited. For dense small aircraft, the gains at both extremely low and high IoU intervals also indicate improved suppression of feature confusion under complex background interference.

TABLE IV
PRECISION OF PROPOSALS IN DIFFERENT SCENARIOS

Scenario	Proposal Precision (no PASTE)	Proposal Precision (with PASTE)
Offshore Ships	81.4%	93.1%
Inshore Ships	56.7%	59.5%
Terminal Aircraft	63.8%	66.6%

The baseline detector is Oriented R-CNN (ResNet-50).

We also compute proposal precision for each scene, defined as the ratio of proposals containing a GT target to the total number of proposals. As shown in Table IV, PASTE yields higher proposal precision in all three scenes.

These qualitative and quantitative results characterize the effect of PASTE on proposal generation in the evaluated scenes.

E. Comparisons With Related Scattering-Aware Methods

Beyond generic baseline detectors, this section evaluates component-level reproductions of two representative scattering-aware methods related to PASTE: the

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TABLE V
COMPARISON WITH SIMILAR METHODS ON FAIR-CSAR

Method	Reproduced Components	SL mAP@0.5	FSI mAP@0.5
SKG-Net [45]	Harris-Laplace Keypoints + Deform Conv + Chamfer Loss	40.4	47.1
ASC-U2Det [47]	ASC Mask Head	41.5	48.2
PASTE (Ours)	ASC-Keypoints + Topology Embedding	**43.3**	**48.3**

Scattering-Keypoint-Guided Network (SKG-Net) [45] and the ASC-Guided Network (ASC-U2Det) [47]. SKG-Net integrates Harris–Laplace keypoint detection with deformable convolution and Chamfer loss, whereas ASC-U2Det uses dense ASC reconstruction masks with subaperture augmentation. Because official open-source implementations are unavailable, only the keypoint-related components described in the original papers are reproduced. The results are summarized in Table V.

As shown in Table V, the PASTE variant reports the highest mAP among the evaluated implementations on both SL and FSI. On SL, it improves over the two component-level reproductions, consistent with the benefit of jointly modeling ASC keypoints and their topology for fine-grained scattering structures. On FSI, however, the performance gap becomes marginal. The smaller gap on FSI may reflect its lower spatial resolution, at which each ASC keypoint covers a larger physical area. Under this condition, sparse keypoints and dense ASC reconstruction masks may provide more similar spatial guidance.

This observation is consistent with the use of ASC-guided physical priors for scattering representation. Under the higher-resolution SL setting, the sparse topology representation yields a larger margin over the evaluated component-level reproductions.

F. Visual Analysis of Predicted Spatial Scattering Map

To further interpret how the spatial scattering map predicted by STIM contributes to detection performance, Fig. 8 visualizes the spatial attention prediction results in three representative scenarios. The left column shows the test samples with GT bounding boxes, while the right column presents the corresponding spatial scattering map predicted by the proposed model.

Specifically, Fig. 8(a) corresponds to the airport scene, where the attention responses are mainly concentrated on the fuselage and wing regions of aircraft while avoiding strong scattering interference from terminal buildings. Fig. 8(b) shows the ground oil-tank scene, in which the model's attention is distributed along the tank boundaries, forming ring-shaped activation patterns aligned with the circular geometry of the storage tanks. Fig. 8(c) presents the near-shore port scene, where the attention is primarily focused on ship contours and dock regions, while background clutter such as buildings and water surfaces exhibits weak activation. These visualizations show that the attention prediction network, supervised by ASC-based keypoints automatically generated through SKAA, captures target geometry and scattering structure in complex SAR scenes.

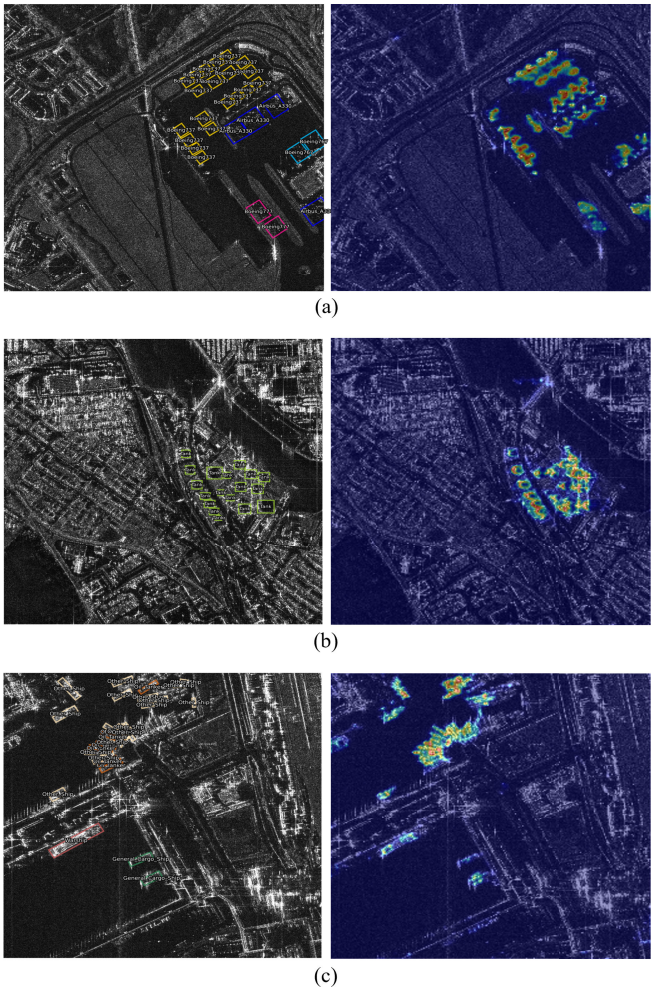


Fig. 8. Visualization of predicted spatial scattering map in different scenarios. (Left column) test samples with GT bounding boxes, while (right column) the corresponding spatial scattering map predicted by the proposed model. From (top to bottom) (a) aircraft detection at an airport, (b) oil tank detection in a ground industrial area, and (c) near-shore ship detection at a port. The highlighted regions in the scattering map correspond to the areas where the prediction network in STIM focuses on the key scattering structures.

Fig. 9 further shows the predicted scattering maps within the target regions. The responses are concentrated near the scattering structures in the displayed examples, and some maps outline portions of the target geometry. These visualizations illustrate the spatial behavior learned by the STIM map-prediction branch.

Together with the ablation results, these visualizations indicate that SKAA supervision guides the predicted maps toward target-associated scattering regions.

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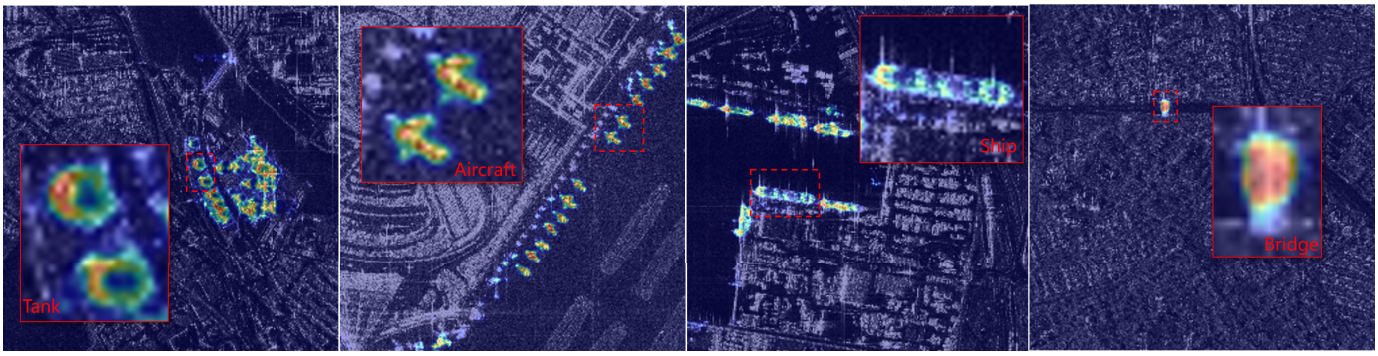


Fig. 9. Predicted spatial scattering map on target region across representative categories.

TABLE VI
ABLATION STUDIES FOR DIFFERENT COMPONENTS OF PASTE

STIM	SPSS	Keypoint Type	mAP@0.5
✗	\	\	38.0
✓	✗	\	39.9
✓	✓	AMP-Keypoints	40.6
✓	✓	SKAA-Keypoints	**42.3**

The baseline detector is RoI Transformer.

G. Ablation Studies of Main Components in PASTE

This subsection examines the contributions of STIM, SPSS, and the SKAA-generated keypoints. The ablation is organized into two sets according to whether SPSS is used. With SPSS enabled, we compare SKAA-Keypoints with amplitude-statistics-based local extrema (amplitude-based keypoints); without SPSS, no keypoint supervision is used.

To compare the ASC-derived keypoints with amplitude-based alternatives, we use a baseline approach that extracts AMP-Keypoints directly from SAR images. Specifically, we employ a Difference-of-Gaussian (DoG)–based detector [53] to obtain salient amplitude extrema. Given an input amplitude image $\boldsymbol{I}$, two Gaussian-smoothed images are generated using standard deviations $\sigma_1 = 1.0$ and $\sigma_2 = 1.6$ with a 3×3 kernel. The DoG response is then computed as

$$\boldsymbol{D} = \mathcal{G}_{\sigma_2}(\boldsymbol{I}) - \mathcal{G}_{\sigma_1}(\boldsymbol{I}). \tag{13}$$

Spatial locations satisfying $|\boldsymbol{D}(x, y)| > 5$ are identified as candidate keypoints, which are further ranked according to their absolute DoG magnitudes. The top-30 points are retained as amplitude-dominant keypoints. Then, these points are clustered into nine representative keypoints for each target. These AMP-Keypoints provide a baseline for comparing image-statistics-based and ASC-based keypoint generation.

RoI Transformer is selected for the ablation study because its learnable spatial transformation and rotation-equivariant features make it sensitive to geometric-topology information, and it records the largest gain with PASTE in Section IV-D. The results in Table VI show that STIM alone, supervised only by the downstream detection loss, yields a limited improvement. Joint optimization of SPSS and STIM improves performance further through the keypoint-based scattering topology prior. The amplitude-based DoG keypoints improve mAP by 0.7 percentage points, whereas the ASC-derived SKAA keypoints improve it by 2.4 percentage points. Thus, ASC positions derived from the electromagnetic parametric representation provide a more effective keypoint prior than positions selected from image amplitude alone.

Furthermore, Fig. 10 compares the two keypoint types under identical SAR crops and target boxes. In the displayed examples, SKAA-Keypoints remain more closely associated with the target extents, whereas AMP-Keypoints are more dispersed toward nearby bright scatterers. Among the displayed crops, 167 of 180 SKAA points (92.8%) lie inside their associated oriented boxes, compared with 154 of 180 AMP points (85.6%). Together with the mAP results in Table VI, these examples indicate that ASC-derived keypoints provide a more target-consistent topology prior in the tested cluttered scenes.

H. Ablation Studies of SKAA and SPSS Hyperparameters

To investigate the influence of the key hyperparameters on detection performance, we conduct ablation studies on the three SKAA settings that determine the generated topology prior, namely the fixed keypoint number k, decoupling threshold τ, and maximum iteration count $N_{\max}$, together with the SPSS weight λ and Gaussian radius σ. Oriented R-CNN is selected as the baseline detector for this section.

1) Effect of SKAA Parameters: Table VII reports an 11-group, single-factor-controlled sensitivity study. The threshold block (T1–T3) varies τ while fixing $N_{\max} = 20$ and $k = 9$; the iteration block (N1–N3) varies $N_{\max}$ while fixing $\tau = -3$ dB and $k = 9$; and the keypoint block (K1–K5) varies k while fixing $\tau = -3$ dB and $N_{\max} = 20$. The default configuration $(\tau, N_{\max}, k) = (-3\ \text{dB}, 20, 9)$ is shown in T2, N2, and K3 as the common control.

For threshold sensitivity, $\tau = -3$ dB obtains 43.30% mAP@0.5, compared with 42.26% at -1 dB and 42.53% at -5 dB. Raising the threshold to -1 dB makes the extracted blocks overly compact, whereas lowering it to -5 dB retains more weak responses. For the iteration sweep, reducing $N_{\max}$ to 10 lowers mAP to 42.84%, consistent with incomplete extraction of secondary structures, while increasing it to 30 lowers mAP to 42.58%, indicating that additional weak

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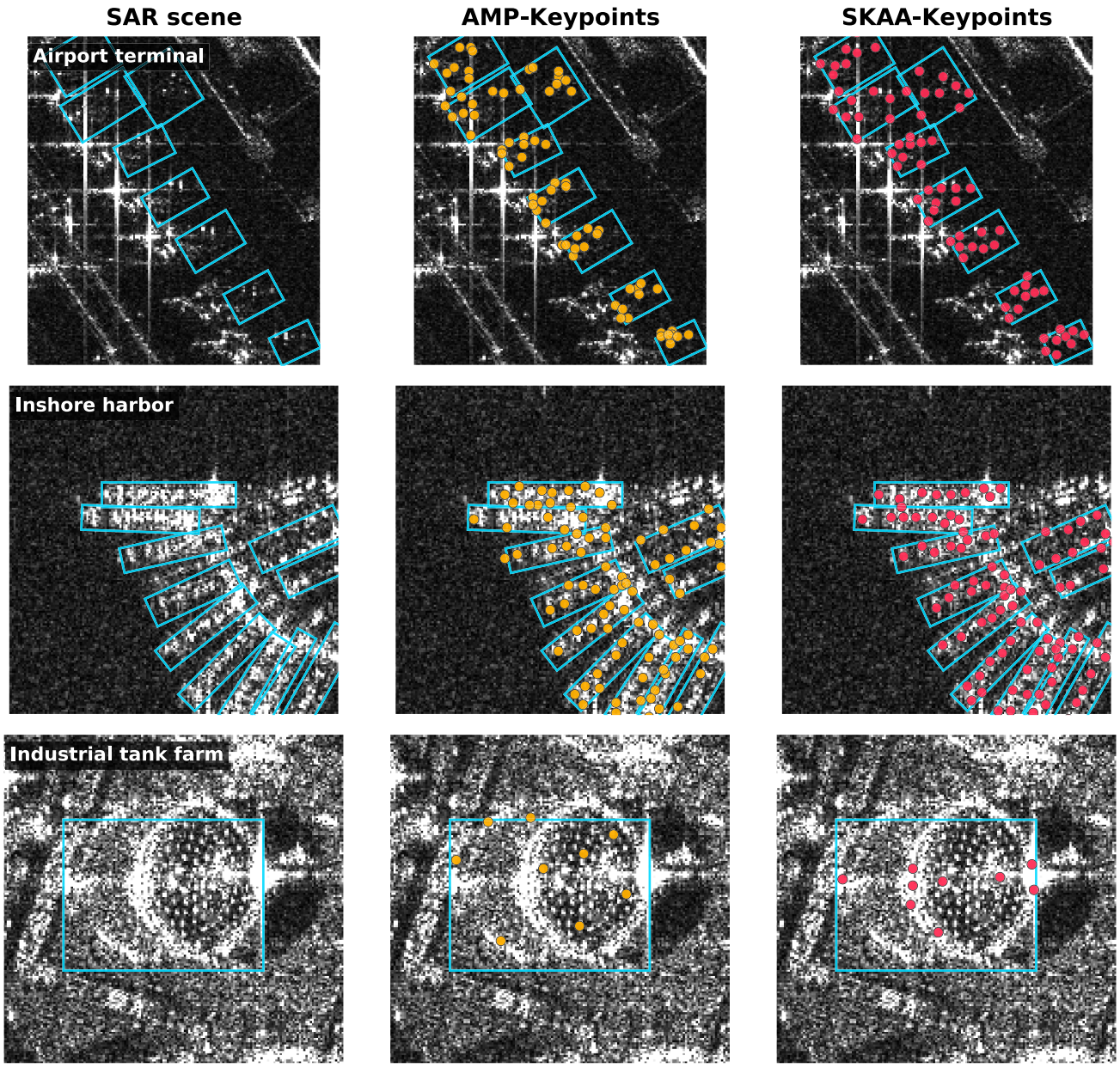


Fig. 10. Comparison of AMP-Keypoints and SKAA-Keypoints in cluttered SAR scenes. The columns show the same SAR crop without keypoints, with AMP-Keypoints (orange), and with SKAA-Keypoints (red), respectively; cyan quadrilaterals denote the associated OBBs. The rows show an airport terminal, an inshore harbor, and an industrial tank farm. Compared with amplitude-based extrema, the ASC-derived keypoints are less dispersed toward strong surrounding clutter and remain more consistently associated with the target scattering structures.

residual regions do not improve the fixed-cardinality topology prior. For the keypoint sweep, mAP first increases and then decreases as k grows: $k = 3$, 6, 9, 12, and 15 achieve 41.60%, 41.40%, 43.30%, 42.60%, and 42.10%, respectively. Within this controlled sensitivity sweep, the default setting provides the highest observed mAP.

2) Effect of SPSS Hyper-Parameters: The results of varying λ and σ are presented in Table VIII. The SPSS-weight ablation keeps the bounding-box regression and classification loss weights fixed at 1 and evaluates $\lambda = 0.1, 0.5, 1$, and 2. The detector achieves the highest mAP when $\lambda = 1$ under this weighting configuration. The numerical loss contributions are not necessarily equal because their magnitudes depend on the individual loss values.

For the Gaussian radius σ, we tested values of 1, 2, 4, and 8. The results show that detection performance decreases as σ increases, from 43.3% at $\sigma = 1$ to lower values. This trend is consistent with the need to preserve localized supervision around the scattering structures. A larger Gaussian radius diffuses the supervision across neighboring pixels and weakens the localization constraint.

I. Robustness Analysis to ASC Localization Error

In practical scenarios, the ASC positions estimated by the SKAA module may inevitably contain localization errors due to noise, model approximation, or imperfect scattering-region decoupling. To evaluate the robustness of the proposed PASTE framework against such perturbations, this section conducts a

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TABLE VII
SINGLE-FACTOR-CONTROLLED SENSITIVITY ANALYSIS OF SKAA PARAMETERS

Ablation Block	Group	τ (dB)	$N_{\max}$	k	mAP@0.5 (%)
Threshold sensitivity	T1	-1	20	9	42.26
	T2 (default)	-3	20	9	**43.30**
	T3	-5	20	9	42.53
Maximum iterations	N1	-3	10	9	42.84
	N2 (default)	-3	20	9	**43.30**
	N3	-3	30	9	42.58
Keypoint number	K1	-3	20	3	41.60
	K2	-3	20	6	41.40
	K3 (default)	-3	20	9	**43.30**
	K4	-3	20	12	42.60
	K5	-3	20	15	42.10

Results are mAP@0.5 (%) on FAIR-CSAR SL using Oriented R-CNN (ResNet-50) with PASTE. Each block varies one factor while retaining the other two at the default $(\tau, N_{\max}, k) = (-3\ \text{dB}, 20, 9)$. The default configuration is listed in T2, N2, and K3 as the common control. Each value is the best validation mAP obtained during the 15-epoch training schedule.

TABLE VIII
ABLATION STUDIES ON SPSS HYPERPARAMETERS

SPSS Weight λ		Gaussian Radius σ	
λ	mAP@0.5	σ	mAP@0.5
0.1	42.5	**1**	**43.3**
0.5	42.2	2	42.4
1	**43.3**	4	42.1
2	41.8	8	41.1

The baseline detector is Oriented R-CNN (ResNet-50).

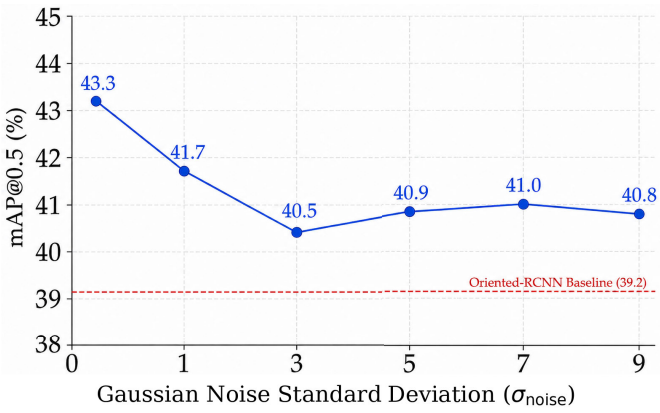


Fig. 11. Robustness analysis of the proposed PASTE framework under different levels of Gaussian perturbation applied to ASC keypoints.

controlled experiment by injecting random Gaussian noise into the ASC keypoint coordinates.

Specifically, let the original ASC keypoint be (x_i, y_i). We perturb its position as

$$\tilde{x}_i = x_i + \epsilon_i^x, \quad \tilde{y}_i = y_i + \epsilon_i^y, \quad \epsilon_i^x, \epsilon_i^y \sim \mathcal{N}\left(0, \sigma_{\text{noise}}^2\right). \tag{14}$$

Here, σ_{noise} denotes the standard deviation in native SAR-image pixels, before any detector-specific resize or augmentation. To state the perturbation relative to image and object scales, for an image of size $H_{\text{img}} \times W_{\text{img}}$ and an oriented target of width w_b and height h_b, we define

$$\rho_{\text{img}} = \frac{\sigma_{\text{noise}}}{\min\left(H_{\text{img}}, W_{\text{img}}\right)}, \qquad \rho_{\text{obj}} = \frac{\sigma_{\text{noise}}}{\sqrt{w_b h_b}}. \tag{15}$$

If the ground sampling distance is r meters per pixel, the corresponding physical standard deviation is $r\sigma_{\text{noise}}$ meters. A subsequent isotropic resize by factor a changes the coordinate standard deviation to $a\sigma_{\text{noise}}$ pixels. We generate noisy annotation sets with $\sigma_{\text{noise}} = 1, 3, 5, 7, 9$ and add them to the ASC keypoint annotations for training, while keeping all other training configurations unchanged. Oriented R-CNN is used as the baseline detector, and the highest mAP@0.5 during training is recorded. The results are shown in Fig. 11.

As shown in Fig. 11, detection performance generally decreases as σ_{noise} increases. The mAP@0.5 drops from 43.3% at $\sigma_{\text{noise}} = 0$ to 40.5% at $\sigma_{\text{noise}} = 3$, while the results at larger perturbations remain around 41% and above the baseline. These results indicate that PASTE retains useful detection performance under the tested ASC localization perturbations.

Across the tested perturbation levels, PASTE remains above the baseline, although its performance is sensitive to moderate localization errors.

J. Computational Overhead

As reported in Table IX, the computational overhead of PASTE comprises offline and online components. SKAA is performed once during preprocessing for each prepared training split and introduces no additional cost during subsequent training or inference. On FAIR-CSAR, the average SKAA time is 413.5 ms per sample; total preprocessing time depends on the number of samples and available parallelism.

For the online stages, the model size increases by 7.1% (from 315.2 to 337.7 MB), and the training time per epoch rises by 12% (from 588.4 to 661.5 s). During inference, the FLOPs increase by 29.7%, and the latency per image increases by 17.3%, mainly because of the scattering-map prediction branch and feature modulation in STIM.

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TABLE IX
COMPUTATIONAL COST AND MEMORY OVERHEAD OF PROPOSED PASTE

Stage	Metric	Baseline	PASTE	Overhead
Annotation Generation	Avg time per sample (ms)	–	413.5	–
Training	Model size (MB)	315.2	337.7	+7.1%
	Training time per epoch (s)	588.4	661.5	+12%
Inference	FLOPs (G)	211.44	274.3	+29.7%
	Latency per image (ms)	61.7	72.4	+17.3%

Overhead is calculated relative to the baseline detector.

The measurements in Table IX are for Oriented R-CNN. Both the absolute and relative overhead depend on the feature-map sizes, channel dimensions, and baseline latency of the detector; detector-specific profiling is therefore needed for other backbones.

V. CONCLUSION

This article proposed PASTE, a closed-loop framework that converts ASC-derived scattering topology into detector-compatible supervision through three coupled components: SKAA automatically generates scattering keypoints, STIM injects their topology into multiscale features, and SPSS aligns the predicted scattering maps with Gaussian soft targets.

Experiments on FAIR-CSAR show that PASTE consistently improves seven detectors spanning single-stage, two-stage, anchor-based, anchor-free, CNN, Transformer, and state-space designs, with relative mAP@0.5 gains of 1.3%–11.3%; ablations and visualizations further confirm the benefit of ASC-derived keypoints, interpretable foreground–background separation, and robustness to ASC localization errors.

The current study is limited by fixed-cardinality keypoints and reliance mainly on ASC positions. Therefore, future work will develop adaptive-cardinality topology representations and incorporate richer electromagnetic attributes.

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